PLASMA PHYSICS & TOKAMAK CONFINEMENT — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: April 16, 2026

Diagram-based questions on Plasma Physics & Tokamak Confinement

Question 1 10.0 points

Examine the toroidal and poloidal magnetic field geometry in the tokamak cross-section depicted in the diagram below. Discuss how these fields collaborate to confine plasma and identify any potential areas of instability within the configuration.

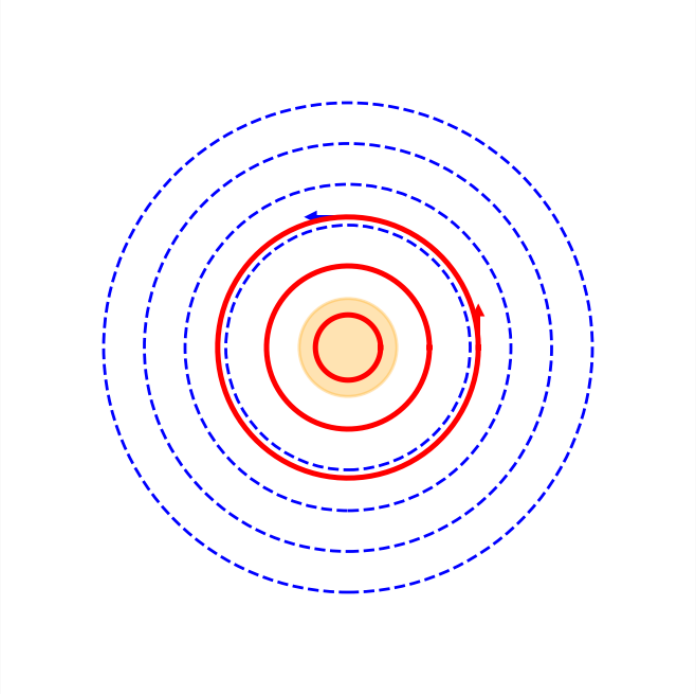

Fig. 1. A cross-sectional diagram of a tokamak showing the toroidal and poloidal magnetic fields. The toroidal field lines should be depicted as concentric circles around the central axis of the tokamak, representing the magnetic field that runs along the torus. The poloidal field lines should be shown as loops encircling the plasma, perpendicular to the toroidal field lines, illustrating the magnetic field that wraps around the plasma vertically. The diagram should label the toroidal field, poloidal field, and plasma region, with arrows indicating the direction of the magnetic fields. Highlight potential areas of instability where the fields might not provide sufficient confinement. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • Together • Instabilities • nash equilibrium • equal areas The diagram will FAIL review if any of these appear.

Question 2 10.0 points

Interpret the Lawson criterion diagram below that displays the triple product $nT\tau$ versus temperature. Explain how this diagram is used to evaluate the feasibility of fusion reactions.

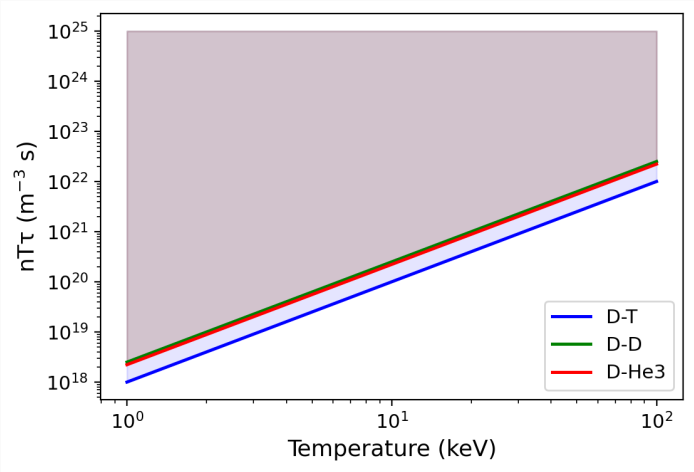

Fig. 2. Lawson criterion plot showing the triple product nTτ versus temperature for different fusion reactions. The diagram includes labeled regions indicating the feasibility of fusion reactions, with axes for temperature (in keV) and the triple product nTτ (in m^-3 s). The plot should highlight regions where different fusion reactions become feasible, such as D-T, D-D, and D-He3, with lines or curves representing the required conditions for energy breakeven and ignition. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • Lawson • Achieving • feasible region • threshold • threshold region • critical threshold The diagram will FAIL review if any of these appear.

Question 3 10.0 points

Examine the diagram below showing magnetic field curvature and grad-B drift for a charged particle in a tokamak. Identify how these drifts affect particle confinement and propose mitigation strategies.

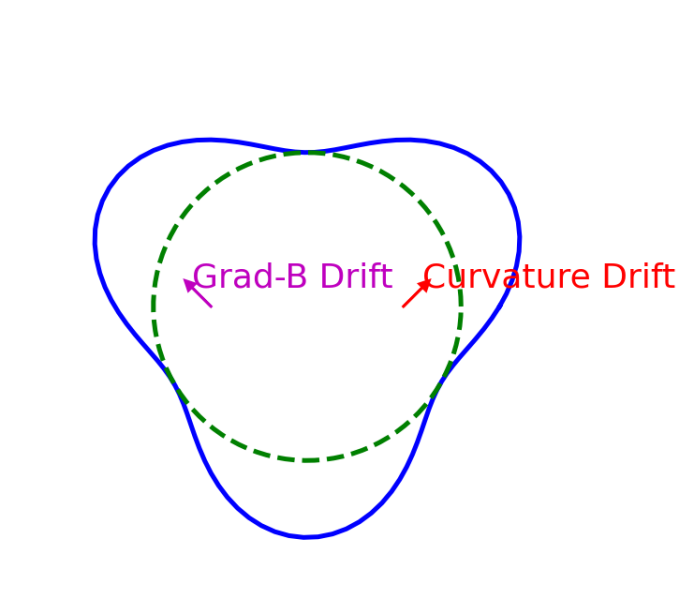

Fig. 3. A diagram showing magnetic field lines in a tokamak with a charged particle experiencing curvature drift and grad-B drift. The diagram includes magnetic field lines curving around the tokamak, with a charged particle shown moving along these lines. The curvature drift is depicted as a perpendicular motion to the magnetic field lines due to the curvature, and the grad-B drift is shown as a drift perpendicular to both the magnetic field and the gradient of the magnetic field strength. Arrows indicate the direction of the drifts, and labels identify the magnetic field lines, curvature drift, and grad-B drift. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • Mitigation The diagram will FAIL review if any of these appear.

Question 4 10.0 points

Examine the q-profile (safety factor) and magnetic flux surfaces in the tokamak depicted in the diagram below. Discuss how the shape of the q-profile influences plasma stability and confinement.

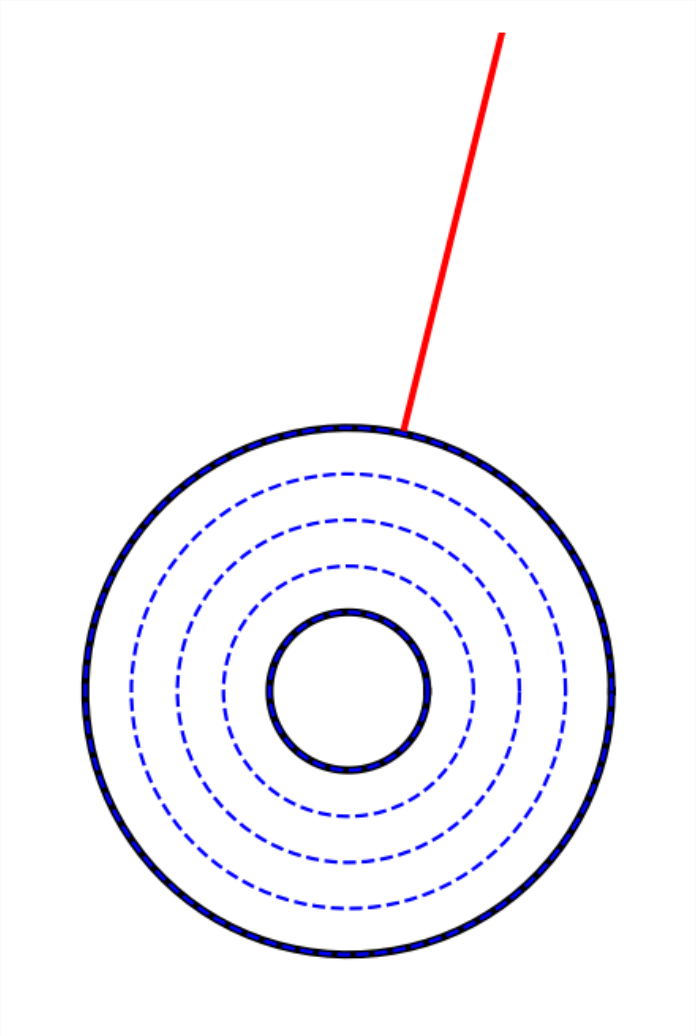

Fig. 4. A diagram of a tokamak showing the magnetic flux surfaces and the q-profile (safety factor). The tokamak should be represented as a toroidal shape, with concentric magnetic flux surfaces depicted inside. The q-profile should be illustrated as a curve or graph overlaying the flux surfaces, indicating how the safety factor varies radially across the plasma. Label the tokamak, magnetic flux surfaces, and q-profile clearly. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • MHD The diagram will FAIL review if any of these appear.

Question 5 10.0 points

Review the diagram below illustrating plasma instability growth rate as a function of wave number. Explain the implications of this relationship for tokamak operation and stability control.

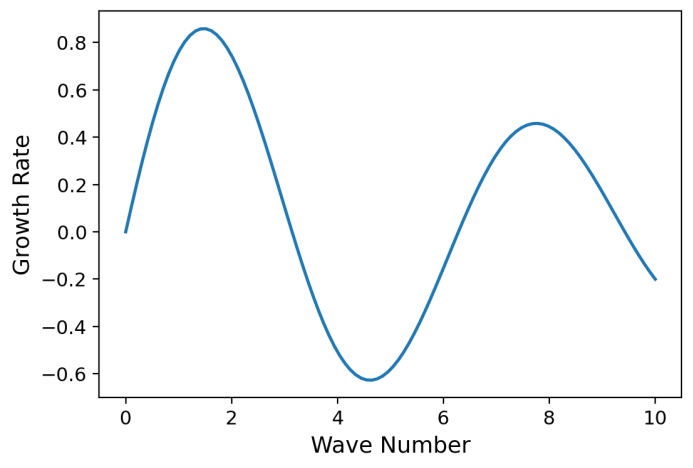

Fig. 5. A plot showing plasma instability growth rate as a function of wave number. The x-axis represents the wave number, and the y-axis represents the growth rate. The plot should illustrate how the growth rate changes with varying wave numbers, which is critical for understanding tokamak operation and stability control. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • Efficient The diagram will FAIL review if any of these appear.